

Modelo de la diapositiva 6/34.

Modelo nuevo dos poblaciones.

Modelo nuevo cuatro poblaciones.

In[104]:=

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Module[{f = 0.49,  $\phi$  = 4.09, pd = 1 / 17.15, v = 0.05,  $\delta$  = 0.139,  $\delta m$  = 25 * 0.0272,
 $\rho$  = 0.1235, pi = 0.0005, K = 55, Di0 = 10, B0 = 60, x30 = 20, x40 = 30, tt = 50,
 $\mu$  = 0.01,  $\alpha$  = 0.01,  $\gamma$  = 0.01,  $\lambda$  = 0.01,  $\beta$  = 0.02,  $\rho s$  = 0.01, as = 0.01, cs = 0.005},
r1 = - $\delta$  * 0.1; b12 = 0.01 *  $\phi$ ; a1 = 1 * as + 0 *  $\phi$  * pd; c1 = cs;
r2 = - $\mu$ ; b21 = -pi * 10; b24 =  $\alpha$ ; a2 = 1 * as + 0 *  $\rho$  / K; c2 = cs;
r3 = - $\gamma$ ; b13 = -v; b31 =  $\lambda$ ; a3 = as; c3 = cs;
r4 =  $\rho s$ ; b42 =  $\beta$ ; a4 = 1 * as + 0 *  $\rho s$  / K; c4 = cs;
eqs = NDSolve[{
(* D'[t] ==  $\phi$  * D[t] * (B[t] - pd * D[t]) - ( $\delta$  + v * P[t]) * D[t] *)
(* B'[t] ==  $\alpha$  * B[t] * S[t] -  $\mu$  * B[t] - pi * B[t] * D[t] *)
(* P'[t] ==  $\lambda$  * P[t] * D[t] -  $\gamma$  * P[t] *)
(* S'[t] ==  $\rho s$  * S[t] * (1 - S[t] / K) -  $\beta$  * B[t] * S[t] *)
(* x1'[t] == x1[t] * (r1 + b12 * x2[t] + b13 * x3[t] - (a1 + c1 * (b12 * x2[t] + b13 * x3[t])) * x1[t]) *)
(* x2'[t] == x2[t] * (r2 + b21 * x1[t] + b24 * x4[t] - (a2 + c2 * (b21 * x1[t] + b24 * x4[t])) * x2[t]) *)
(* x3'[t] == x3[t] * (r3 + b31 * x1[t] - (a3 + c3 * b31 * x1[t]) * x3[t]) *)
(* x4'[t] == x4[t] * (r4 + b42 * x2[t] - (a4 + c4 * b42 * x2[t]) * x4[t]) *)
x1[0] == Di0,
x2[0] == B0,
x3[0] == x30,
x4[0] == x40},
{x1, x2, x3, x4}, {t, 0, tt}, Method -> {"TimeIntegration" -> "ExplicitRungeKutta"}];

Print@
TableForm[{{"", "x1", "x2", "x3", "x4"}, {"ri", r1, r2, r3, r4}, {"ai", a1, a2, a3, a4},
{"ci", c1, c2, c3, c4}, {"", "b12", "b21", "b13", "b31", "b24", "b42"},
{"bij", b12, b21, b13, b31, b24, b42}}];
Plot[Evaluate[{x1[t], x2[t], x3[t], x4[t]} /. eqs],
{t, 0, tt}, PlotLegends -> {"D", "B", "P", "S"}, PlotRange -> All]]
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|     | x1      | x2     | x3    | x4    |      |      |
|-----|---------|--------|-------|-------|------|------|
| ri  | -0.0139 | -0.01  | -0.01 | 0.01  |      |      |
| ai  | 0.01    | 0.01   | 0.01  | 0.01  |      |      |
| ci  | 0.005   | 0.005  | 0.005 | 0.005 |      |      |
|     | b12     | b21    | b13   | b31   | b24  | b42  |
| bij | 0.0409  | -0.005 | -0.05 | 0.01  | 0.01 | 0.02 |

Out[104]=

